

HARSHIL SHARMA

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EDUCATION

SYMBIOSIS INSTITUTE OF TECHNOLOGY

Bachelor of Technology

Major in Computer Science; Specialization in Data Science

Pune, India

2020 - 2024

Relevant Coursework: Data Science, Business Analytics, Data Warehousing & Mining, Big Data, HCI (Human Computer Interface), Machine Learning, Database Management Systems, Cyber Security, Cloud Computing Tools & Techniques

WORK EXPERIENCE

FRACTAL ANALYTICS

Data Scientist

Bangalore, India

Sept 2024 – Present

Client - Carrier (TariffIQ GenAI Assessment- Tariff Regulation Interpretation and Financial Exposure Analysis using AWS Multi-Agent Architecture)

- Contributed across assessment and pilot phases of TariffIQ, mapping regulatory changes to product-market categories and quantifying P&L and margin exposure to support data-driven sourcing decisions.
- Developed the Trade Regulation Interpretation Agent on AWS (Bedrock, Lambda, RDS), automating HS code classification via USITC API and benchmarking LLMs (Claude, Nova Pro, Cohere) against a client-validated golden dataset.

Client - BHP Australia (Behavioural Data Science for Workplace Safety)

- Led a GenAI-driven Behavioural Data Science initiative for BHP to reduce workplace fatalities using data-driven insights.
- Partnered with neuroscience and brain science experts to uncover cognitive and behavioural factors behind safety incidents.
- Designing a GenAI-powered solution combining advanced analytics, behavioural modeling, and generative AI to tackle critical safety challenges.

Client - Carrier (Automation of Distributor Receipt Order Processing using AWS GenAI Vision Models)

- Collaborated within a cross-functional team to design and prototype an AI-driven order automation system using Generative Vision-Language Models (AWS Nova, Claude, Mistral).
- Developed a GenAI pipeline to extract structured data from multi-line PDF receipts and automate CSV exports to SAP.
- Conducted model experimentation and API integration under AWS RAPID program, applying prompt engineering, RAG, and embedding-based retrieval.
- Ensured data security, PII compliance, and scalable GenAI adoption in production-ready workflows on AWS.

Client - Molina Healthcare (Full-Stack POC with Agentic AI/LLM Agents)

- Built and deployed backend components for a full-stack POC, gaining hands-on experience with Agentic AI architectures and LLM-based agents.
- Mentored an intern by providing project context and technical guidance, enabling effective contributions.
- Optimized code for internal projects, improving computational efficiency and reducing latency.
- Performed testing and debugging, enhancing accuracy, efficiency, and reliability of AI systems and data workflows.

Product - Cogentiq AI (Multi-Agent Orchestration and LLM Automation)

- Built multi-agent orchestration workflows with Python, LangGraph and Crew AI, improving task routing and reasoning.
- Deployed LLMs with custom prompts for structured claim validation and API-driven automation.

Product - Flyfish AI (Agentic AI and RAG Pipelines for Customer Service Automation)

- Contributed to the development of Agentic AI architectures with LangChain and multi-agent orchestration for intelligent customer service.
- Implemented RAG pipelines integrating ChromaDB with Elasticsearch/Lucene for context-aware LLM responses.
- Optimized bot architecture, LLM parameters, SQL integration, and backend systems for production environments.
- Enhanced pipeline reliability and analytics workflows via log analysis, Python ETL, and low-code automation (Power Automate).

C-DOT (Centre For Development Of Telematics)

Delhi, India

Data Science Intern

Jan 2024 – June 2024

- Developed a **biometric face recognition system** using deep learning, data curation, and image super-resolution techniques.
- Researched global meteorological departments to develop a **Multimodal Disaster Management Dashboard** in Power BI.
- Integrated **social media data** for crisis information extraction and classification; enhanced dashboards for emergency response using Python and web scraping.

KG4ASTRA Extension Programme

Remote

Data Engineering Researcher

June 2023 – Dec 2023

- Built **ontology** and **hierarchy graphs** using industry stand tools like **Neo4j** and **Protege5.5** for classifying Indian missiles , providing a structured and comprehensive understanding of their relationships and attributes.
- Collaborated with a team of experts for developing a **custom NLP model** tailored for segmenting Indian missiles based on hierarchy.

PUBLICATIONS

“SneakerXchange: Enhancing User Experience through HCI Principles in a Reselling Website.”

9th International Conference on Innovative Computing & Communication (ICICC 2026)

Springer Lecture Notes in Networks and Systems | Accepted (Top 15%)

ACADEMIC PROJECTS

Asteroid Hazard Classification ([link](#))

Nov 2023

- Utilized a curated JPL-Caltech (NASA) dataset to analyze diverse astronomical properties using regression algorithms.
- Developed a robust model to classify hazardous vs. non-hazardous asteroids, supporting planetary defense and risk mitigation.

“Kitna Kharcha” - An Innovative approach to expense analytics ([link](#))

Nov 2023

- Developed a predictive urban financial model using data analysis to tackle metropolitan living complexities.
- Implemented a model predicting location-specific costs for housing, transportation, groceries, and leisure, enabling precise financial planning.

Post Pandemic New York Airbnb Analysis - R Programming ([link](#))

Nov 2022

- Analyzed 100K+ Airbnb listings across NYC's five boroughs to study post-Covid pricing recovery patterns, room type distributions, and neighbourhood demand shifts.
- Identified 35% price increase in outer boroughs compared to pre-pandemic levels through statistical modeling and geospatial visualization.

Climate Change Forecasting - Python ([link](#))

Oct 2022

- Implemented Regression based Climate Change Forecasting , analyzing 250 years of historical earth surface temp data
- Developed a Predictive Model for forecasting the average temperature rise by 2050

SneakerXchange- Enhancing User Experience through HCI Principles in a Reselling Website ([link](#))

Oct 2022

- Designed and developed a fully functional website prototype in Figma, applying core Human-Computer Interaction principles to enhance user experience.
- Implemented user-centric features such as clear navigation, intuitive search, personalized recommendations, and mobile responsiveness.

Technical Skills

Programming Languages: Python, R, SQL

Frameworks & Libraries: TensorFlow, PyTorch, Keras, Pandas, NumPy, OpenCV, BeautifulSoup, Playwright, Scrapy, Matplotlib, Seaborn, scikit-learn, ggplot, dplyr, tidyr, caret, ResNet, MobileNet, altair

Machine Learning & Deep Learning: CNN, YOLO, GANs, SVM, Decision Trees, Random Forests, Regression, Time Series, Clustering, NLP, GLM, Segmentation/Cluster Analysis

Generative & Agentic AI: LLMs, VLMs (GPT, Claude, Mistral, Gemini, Nova), RAG Pipelines, Multi-Agent Systems, Prompt Engineering, Guardrails & Hallucination Control

Tools & Platforms: AWS, Power BI, Tableau, CI/CD, Docker, GitLab, Git, Power Automate, Postman, Canva, Figma

Databases & APIs: MongoDB, FAISS, Elasticsearch, ChromaDB, REST APIs, Pydantic, JSON

MLOps / LLMOps: Model evaluation, latency optimization, observability, tracing

Volunteering

DefEating Disorders - New York

Remote

- Created content focused on eating disorder awareness, recovery resources, and community education.
- Developed visual communication materials and social media graphics to support ongoing advocacy initiatives.

Kshamata - Mumbai

Remote

- Wrote blogs, awareness articles, and social media captions supporting advocacy for human-trafficking survivors.
- Designed digital creatives and campaign posts used across Instagram and other outreach platforms

Extra Curricular Activities

Core Team Member - Symbiosis Institute of Technology

Pune, India

- Active in TED Club, Dance Club, Varsity, Photography Club, EBSB, Mosaic Club, and CodeChef.
- Participated in athletics (marathons, cycling), arts (painting, crafts, dancing), and culinary projects.
- Won multiple competitions in general knowledge, art, and dance at school and college level.
- Self-taught cook and baker, demonstrating creativity and discipline.